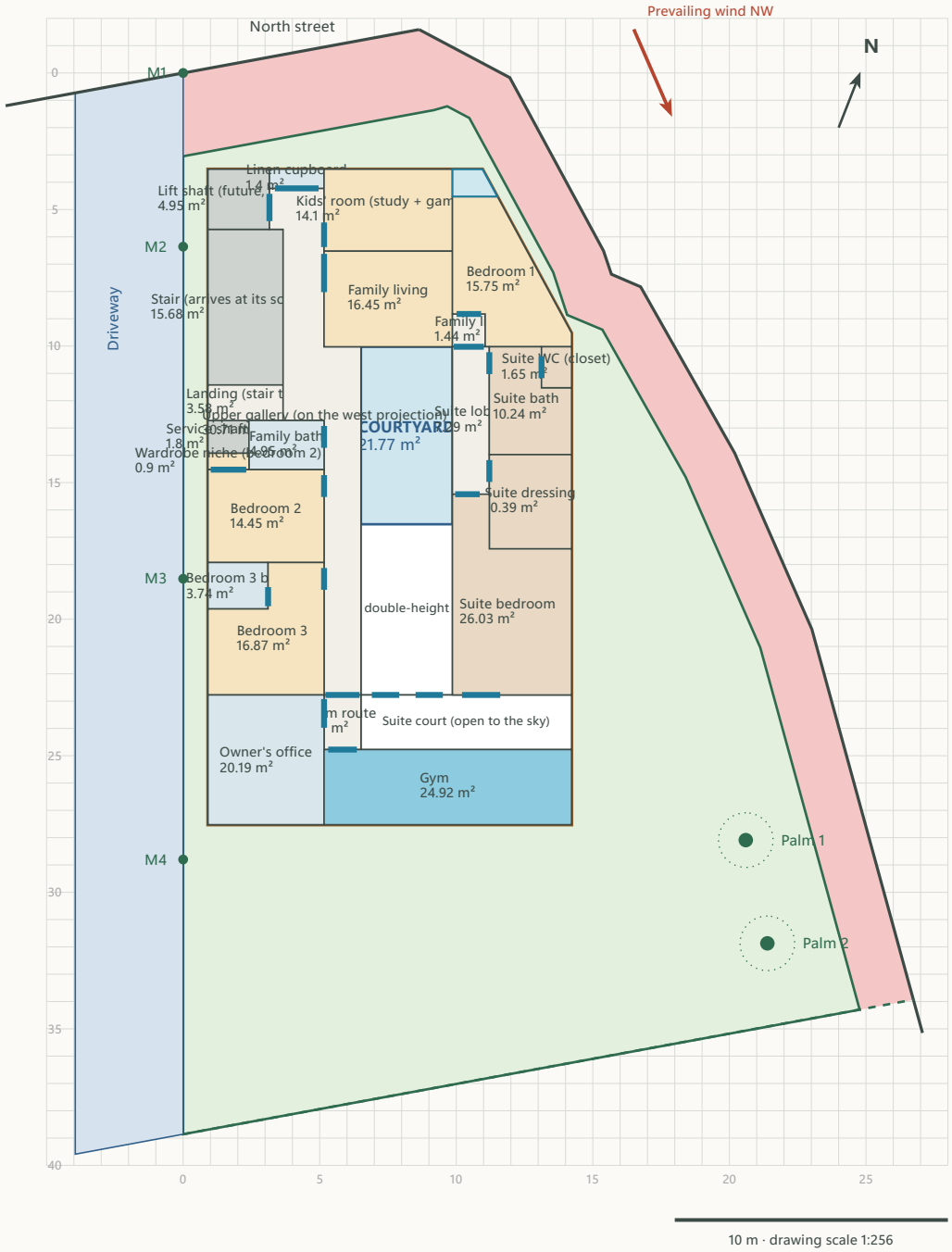



# Taif / Palm House

## First floor V5.0 · family part + the L (office, gym, suite court, master suite)

20 Oct 2026 · on ground floor V5.0 · rooms in data/rooms_ff.csv · design and furniture in studies/ff_family.typ and studies/ff_suite.typ · schedule and light check on the next pages

**First floor V5.0.** The gallery runs on a **1.35 m projection** along the west edge of the courtyard and the double-height void (x 4.25–5.6, edge beam on three columns), so the west rooms take the full 4.25 m band: **family bath**, **bedroom 2**, **bedroom 3 with its own bath**. North band: **kids' room (study + gaming)** and the **family living** on the courtyard edge. **Bedroom 1** is one room on the straight y 6.5 line. The **L**: **owner's office** (west), **gym route**, **gym**, the **suite court** open to the sky in front of the clerestory windows, and the **master suite** (lobby, luxury bath with WC closet, walk-in dressing, bedroom). The east duct is removed. Colours: suite (beige) · family (yellow) · baths, office and linen (blue-grey) · gym (blue) · circulation (light) · fixed (grey) · courtyard, void and court (white / blue). Blue marks: doors and the two clerestory windows. Areas: available 243.17 m² · rooms 227.75 m² (circulation 45.72 m²) · suite court 15.42 m². ✓ **data/rooms_ff.csv matches the two studies room for room.**

✓ **All checks passed**

## Areas

| | |
|---|---|
| Envelope | **667.87 m²** |
| Building limit (envelope unless set) | **309.68 m²** |
| Other open voids | **36.36 m²** |
| Courtyard (open to sky) | **21.77 m²** |
| Rooms placed (23) | **250.18 m²** |
| Limit minus courtyard and voids | **251.55 m²** |
| Air channel + duct | **1.37 m²** |
| Not yet assigned | **1.37 m²** |

## Orientation

| | |
|---|---|
| True compass vs the M line | **21.4°** |
| NW wind vs the M line | **23.6°** |

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

# Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

**Envelope corners**

| Corner | x | y |
|---|---|---|
| V1 | 0 | 38.86 |
| V2 | 0 | 3.05 |
| V3 | 9.18 | 1.36 |
| V4 | 9.67 | 1.22 |
| V5 | 10.48 | 1.65 |
| V6 | 13.55 | 7.3 |
| V7 | 14.06 | 8.86 |
| V8 | 15.35 | 9.4 |
| V9 | 18.4 | 14.8 |
| V10 | 21.13 | 21.04 |
| V11 | 24.77 | 34.3 |

**Rooms**

| Name | Frame | w × d (m) | Area |
|---|---|---|---|
| Stair (arrives at its south end) | M | 2.75 × 5.7 | 15.68 m² |
| Lift shaft (future, 2 stops) | M | 2.25 × 2.2 | 4.95 m² |
| Service shaft (west) | M | 1.5 × 1.2 | 1.8 m² |
| Landing (stair top) | M | 2.75 × 1.3 | 3.58 m² |
| Upper gallery (on the west projection) | M | 3.35 × 18.55 | 30.71 m² |
| Linen cupboard | M | 2 × 0.7 | 1.4 m² |
| Family bath | M | 2.75 × 1.8 | 4.95 m² |
| Wardrobe niche (bedroom 2) | M | 1.5 × 0.6 | 0.9 m² |
| Bedroom 2 | M | 4.25 × 3.4 | 14.45 m² |
| Bedroom 3 bath | M | 2.2 × 1.7 | 3.74 m² |
| Bedroom 3 | M | 4.25 × 4.85 | 16.87 m² |
| Kids' room (study + gaming) | M | 4.7 × 3 | 14.1 m² |
| Family living | M | 4.7 × 3.5 | 16.45 m² |
| Family link | M | 1.2 × 1.2 | 1.44 m² |
| Bedroom 1 | M | 4.361 × 5.5 | 15.75 m² |
| Gym route | M | 1.35 × 2 | 2.7 m² |
| Owner's office | M | 4.25 × 4.75 | 20.19 m² |
| Gym | M | 9.061 × 2.75 | 24.92 m² |
| Suite lobby | M | 1.35 × 5.4 | 7.29 m² |
| Suite bath | M | 3.011 × 3.95 | 10.24 m² |
| Suite WC (closet) | M | 1.1 × 1.5 | 1.65 m² |
| Suite dressing | M | 3.011 × 3.45 | 10.39 m² |
| Suite bedroom | M | 4.361 × 7.35 | 26.03 m² |

# First floor V5.0 · room schedule

wall centre lines (axes) · clear ≈ axes minus half a wall on every edge (from the studies) · CSV coordinates

| Room | Part | Axes w × d (m) | Axes m² | ≈ clear m² |
|---|---|---|---|---|
| Landing (stair top) | family | 2.75 × 1.3 | 3.57 | 3.14 |
| Upper gallery (on the west projection) | family | polygon | 30.71 | 27.65 |
| Linen cupboard | family | 2 × 0.7 | 1.4 | – |
| Family bath | family | 2.75 × 1.8 | 4.95 | 4.04 |
| Wardrobe niche (bedroom 2) | family | 1.5 × 0.6 | 0.9 | – |
| Bedroom 2 | family | 4.25 × 3.4 | 14.45 | 12.84 |
| Bedroom 3 bath | family | 2.2 × 1.7 | 3.74 | 2.92 |
| Bedroom 3 | family | polygon | 16.87 | 14.97 |
| Kids' room (study + gaming) | family | 4.7 × 3 | 14.1 | 12.44 |
| Family living | family | 4.7 × 3.5 | 16.45 | 15.16 |
| Family link | family | 1.2 × 1.2 | 1.44 | 1.08 |
| Bedroom 1 | family | polygon | 15.75 | 13.83 |
| Gym route | L | 1.35 × 2 | 2.7 | 2.03 |
| Owner's office | L | 4.25 × 4.75 | 20.19 | 18.16 |
| Gym | L | 9.06 × 2.75 | 24.92 | 22.26 |
| Suite court (open to the sky) | L | 7.71 × 2 | 15.42 | – |
| Suite lobby | L | 1.35 × 5.4 | 7.29 | 5.94 |
| Suite bath | L | polygon | 10.24 | 8.79 |
| Suite WC (closet) | L | 1.1 × 1.5 | 1.65 | 1.09 |
| Suite dressing | L | 3.01 × 3.45 | 10.39 | 9.01 |
| Suite bedroom | L | polygon | 26.03 | 23.56 |
| **Rooms (indoor)** | | | **227.75** | |
| **Suite court (open)** | | | **15.42** | |
| Stair, lift, west shaft, air tower | | | 23.8 | |

## Key numbers

- **Master suite:** 55.6 m² indoors (lobby, bath + WC closet, dressing, bedroom), + the suite court 15.42 m².
- **Circulation:** 45.72 m² (landing, gallery incl. the projection, family link, gym route, suite lobby).
- **Courtyard sky** 3.35 × 6.5 m and **double-height void** 3.35 × 6.25 m on axes (both were 4.70 m wide before the projection); the ground-floor courtyard and living keep their full floor.
- **Bedrooms 2 and 3:** 4.25 m wide on axes (≈ 4.0 m clear).

## Where the detail is

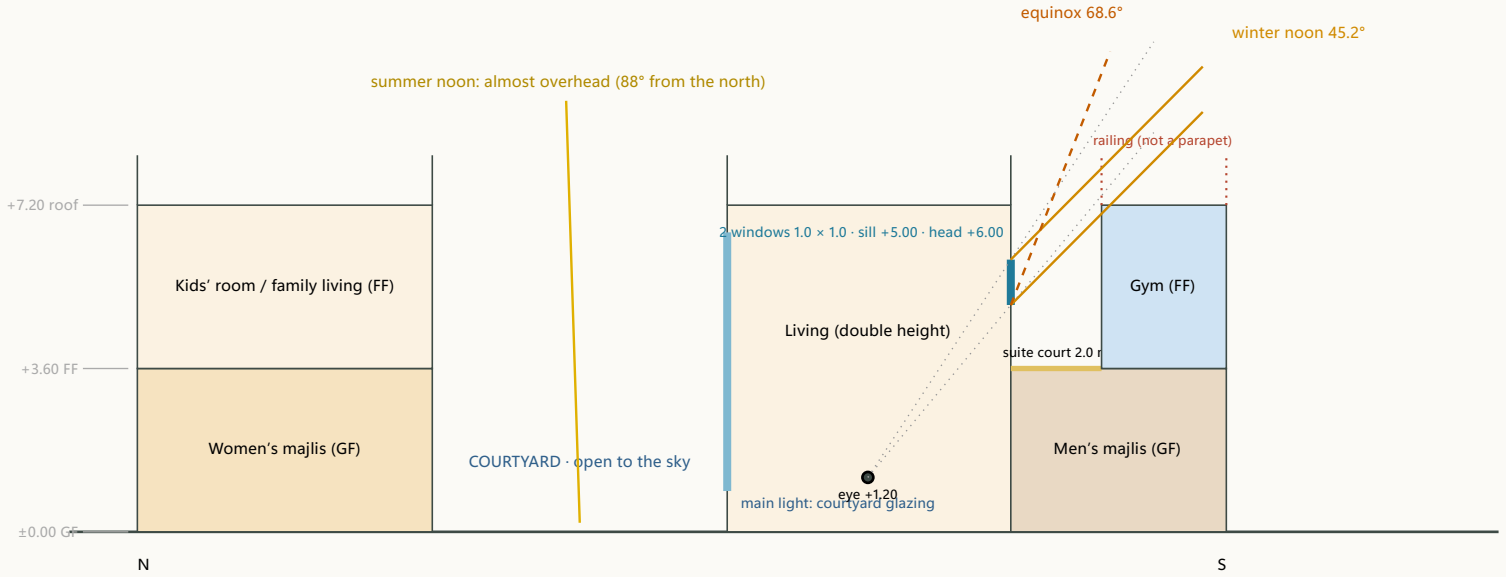
- **Furniture, doors, windows, drains, walking distances, routes ≥ 1.0 m and room checks:** studies/ff_family.typ (part 1) and studies/ff_suite.typ (part 2).
- **Construction notes** (projection, columns, east duct, sunken baths, suite court, cupboards, loads): docs/DETAILS.md.
- **Decisions and open items:** docs/FIRST_FLOOR_BRIEF.md (V5.0) and docs/notes.md.

## Open items

- Engineer: the west projection (columns + edge beam, or a plain cantilever), loads and vibration of the gym and walls over the men’s majlis, the y 3.0 partition over the women’s majlis.
- MEP: the stack outside the east wall (suite bath, vent stack), the kitchen extract through the east wall, the hood position south of y ≈ 7.5.
- The clerestory sill vs the suite court floor (eye height, or obscured glass); a daylight check of the narrower courtyard.

# First floor V5.0 · light check · the double-height living's south windows

section north → south through the middle band (CSV x ≈ 7.3) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.10 · a full daylight simulation comes later




| Behind the 2.0 m suite court | Winter noon sun on the windows | Equinox noon sun | Sky seen through the windows from the eye |
|---|---|---|---|
| gym with a 1.10 m parapet (top +8.30) | 0 % | 100 % | 42 % |
| gym with a railing only (solid top +7.20) · **V5.0** | 81 % | 100 % | 100 % |

**How the percentages are defined.** "Sun on the windows": the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox), in this section; the ray from each point of the window must pass over the top of the gym's north wall 2.0 m away. "Sky seen": from a seated eye 1.20 m high in the middle of the double-height living floor (CSV y 16.1), the share of the vertical angle of the windows that shows sky rather than the gym. Pure geometry: no reflections, no sky model.

- **The main light of the double-height living comes from the courtyard.** The two small south windows (1.0 × 1.0 m, sill +5.00) add light and let warm air out high on the south.
- **V5.0:** the 2.0 m strip in front of them is now the **suite court**, walled on four sides and open to the sky; the gym's north wall is its south side, ending in a **railing, not a parapet**.
- **Privacy:** the windows open onto the suite court, so their sill stays above eye level from the court floor (or obscured glass); to check against the floor levels.
- **Summer:** the sun is almost overhead; light-coloured court walls and small hoods (≈ 0.4 m) over the windows.